Introduction of Research of Language Media Laboratory

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Abstract

I'd like to introduce our research of language media laboratory of Hokkaido University in Japan. Our laboratory focuses on research of languages. The goal of our research is practical realization and its application of linguistic capability on engineering like an adult. At first, we try to realize innate capability of language acquisition like an infant in order to achieve this goal.

Our research projects are very wide on natural language processing. First, we research on language acquisition. These topics are "A Spoken Dialogue Method Using Inductive Learning and Genetic Algorithm with Sexual Selection" and "An Acquisition System of Noun Concepts from an Images and Utterance for them." Second, we research on machine translation. These topics are "An Improvement of Results of Machine Translation Using Web Resources" and "Multilingual Software Translation System Using a Pivot Language." Third, we research on dialogue processing. These topics are "A Chat System Using the Web", "A Knowledge Acquisition System Using Dialogue Media" and "Automatic Recognition and Generation of Humor." Fourth, we research on automatic proofreading. These topics are "A Method of Automatically Detecting and Correcting Article Errors in English Using Inductive Learning" and "A Method of Automatically Detecting and Correcting Postpositional Particle Errors in Japanese Using Inductive Learning." Other research projects are text mining, acquition action of a robot and knowledge acquisition using the Web. These topics are "An Unsupervised Opinion Mining Approach for Japanese Weblog Reputation Information", "Connectives Acquisition in a Humanoid Robot Using Direct Physical Feedback" and "Automatic Extraction of Information of Common Sense and Emotion Using the Web."

I will explain two topics of language acquisition in detail at this talk. At this talk, I will explain a Spoken Dialogue Method Using Inductive Learning Method Based on Genetic Algorithm with Sexual Selection (SeGA-ILSD) along with analysis of its generality for different languages: Japanese and English. This method achieves high generality using a learning function. Therefore, this method can show equivalent

performance for different languages. To confirm it, we developed an experimental system of this method and carried out an experiment of performance evaluation of this method. These results show that this method has equivalent performance for two languages: Japanese and English. Results verify that this proposed method has high generality for these languages.

Next I will explain an Acquisition System of Noun Concepts from an Images and Utterance for them in detail. In case of a robot working in an open environment, the task-oriented language ability will not be sufficient. To adapt to the environment, it is necessary to acquire language dynamically. Therefore we propose a system that acquires noun concepts (labels for images) based on infant vocabulary acquisition models. The experimental results show that the proposed system can acquire noun concepts and is robust against speech recognition errors. We also evaluate user impressions of the system using the rating scale method. The experimental results show that the system leaves a good impression on people.

Next I will explain a Method of Automatically Detecting and Correcting Article Errors in English Using Inductive Learning in detail. One of the most common mistakes made by Japanese learners of English is an article error. We propose a method for correcting and recognizing article errors using features of word appearance. By acquiring rules from word appearances in a corpus and by using learning methods, the proposed system corrects and recognizes article errors with context consideration. Experiments show that the proposed method achieves a F-measure of 0.41, which is better than results of related works (0.35). At last, I will introduce other research shortly.